

SKILL-5: SHELL QUOTING MECHANISMS (SINGLE & DOUBLE QUOTES)

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Task 1 — Single Quotes in Shell

Objective: To apply single quotes in Bash and understand that everything inside `'...'` is treated literally, variable expansion does not happen, spaces are preserved, special characters are treated as normal characters, the quoted string can be stored in a variable, and parsing can be validated using `echo`, `printf`, and `set -x`.

Key Steps & Execution Breakdown:

- **Step 1-3 (Variables & Single Quotes):** Created variable `name="Saishashank"`. Running `echo $name` outputs `Saishashank`, whereas `echo '$name'` outputs literally `$name` because single quotes prevent variable expansion.
- **Step 4-5 (Literal Content & Spaces):** Expressions like `echo 'Hello $name'` and `echo 'Hello * ? [ ] { }'` preserve special characters as normal text. Spaces inside single quotes are preserved without word splitting.
- **Step 6-8 (Assignment & Validation):** Stored quoted strings in variables (e.g., `message="Welcome to Ubuntu Linux"`). Used Bash tracing with `set -x` to inspect how Bash executes quoted commands.
- **Step 9 (Edge Cases):** Tested empty quotes, numbers/symbols, command substitution (`'Today is $(date)'` which remains literal), and handling nested quotes.

Execution Screenshots (Task 1):

```
lenovo@SAISHASHANK:~$ pwd
/home/lenovo
lenovo@SAISHASHANK:~$ /home/saishashank
-bash: /home/saishashank: No such file or directory
lenovo@SAISHASHANK:~$ /home/lenovo
-bash: /home/lenovo: Is a directory
lenovo@SAISHASHANK:~$ name="Saishashank"
lenovo@SAISHASHANK:~$ echo $name
Saishashank
lenovo@SAISHASHANK:~$ echo '$name'
$name
lenovo@SAISHASHANK:~$ $name
lenovo@SAISHASHANK:~$ $name
Saishashank: command not found
lenovo@SAISHASHANK:~$ echo 'Hello $name'
Hello $name
lenovo@SAISHASHANK:~$ Hello $name
Command 'Hello' not found, did you mean:
command 'hello' from snap hello (2.10)
command 'hello' from deb hello (2.10-5build1)
command 'hello' from deb hello (1.6.1-1)
See 'snap info <snapname>' for additional versions.
lenovo@SAISHASHANK:~$ echo 'Hello * ? [ ] { }'
Hello * ? [ ] { }
lenovo@SAISHASHANK:~$ echo 'Hello          Ubuntu          World'
Hello          Ubuntu          World
lenovo@SAISHASHANK:~$ message='Welcome to Ubuntu Linux'
lenovo@SAISHASHANK:~$ echo "$message"
Welcome to Ubuntu Linux
lenovo@SAISHASHANK:~$ printf '%s\n' "$message"
Welcome to Ubuntu Linux
lenovo@SAISHASHANK:~$ message='$name is learning Ubuntu'
```

```
lenovo@SAISHASHANK:~$ echo "Special characters: $ @ # % & *"
Special characters: $ @ # % & *
lenovo@SAISHASHANK:~$ name="Saishashank"
lenovo@SAISHASHANK:~$ echo "$name"
Saishashank
lenovo@SAISHASHANK:~$ printf '%s\n' "$name"
<Saishashank>
lenovo@SAISHASHANK:~$ empty=""
lenovo@SAISHASHANK:~$ printf '%s\n' "$empty"
<>
lenovo@SAISHASHANK:~$ mkdir "My Ubuntu Project"
lenovo@SAISHASHANK:~$ cd "My Ubuntu Project"
lenovo@SAISHASHANK:~/My Ubuntu Project$ pwd
/home/lenovo/My Ubuntu Project
lenovo@SAISHASHANK:~/My Ubuntu Project$ cd
lenovo@SAISHASHANK:~$ rmdir "My Ubuntu Project"
lenovo@SAISHASHANK:~$ nano quotes.sh
lenovo@SAISHASHANK:~$ chmod +x quotes.sh
lenovo@SAISHASHANK:~$ ./quotes.sh
----- SINGLE QUOTES -----
$name
Welcome to Ubuntu
Today is $(date)
Hello          Ubuntu          Linux
----- DOUBLE QUOTES -----
My name is Aruntej
I am studying Computer Science
Current directory: /home/lenovo
Current date: Tue Sep 22 04:38:53 UTC 2026

With double quotes:
<Ubuntu Linux Lab>

Without double quotes:
<Ubuntu>
<Linux>
<Lab>
lenovo@SAISHASHANK:~$
```


Task 2 — Double Quotes in Shell

Objective: To apply double quotes and understand that spaces are preserved, variables are expanded, command substitution works, nested tokens can be handled, quoted commands can be tested, and parsing can be validated.

Key Steps & Execution Breakdown:

- **Step 1-4 (Variable Expansion & Spaces):** Defined name="Saishashank" and course="Computer Science". Double quotes allow variable expansion (e.g., echo "My name is \$name") while preserving spaces and preventing unwanted word splitting compared to unquoted variables.
- **Step 5-8 (Command Substitution & Nested Tokens):** Executed command substitution like echo "Current date: \$(date)" and echo "Files in current directory: \$(ls)". Handled directories with spaces such as mkdir "My Ubuntu Project" and navigating with cd.
- **Step 9-14 (Validation & Edge Cases):** Used printf '<%s>\n' "\$text" to prove quoted variables are treated as a single argument, tested empty variables (empty=""), and validated parsing using Bash tracing (set -x).

Execution Screenshots (Task 2):

```
lenovo@SAISHASHANK:~$ echo "Special characters: $ @ # % & *"
Special characters:  @ # % & *
lenovo@SAISHASHANK:~$ name="Saishashank"
lenovo@SAISHASHANK:~$ echo "$name"
Saishashank
lenovo@SAISHASHANK:~$ printf '<%s>\n' "$name"
<Saishashank>
lenovo@SAISHASHANK:~$ empty=""
lenovo@SAISHASHANK:~$ printf '<%s>\n' "$empty"
<>
lenovo@SAISHASHANK:~$ mkdir "My Ubuntu Project"
lenovo@SAISHASHANK:~$ cd "My Ubuntu Project"
lenovo@SAISHASHANK:~/My Ubuntu Project$ pwd
/home/lenovo/My Ubuntu Project
lenovo@SAISHASHANK:~/My Ubuntu Project$ cd ..
lenovo@SAISHASHANK:~$ rmdir "My Ubuntu Project"
lenovo@SAISHASHANK:~$ nano quotes.sh
lenovo@SAISHASHANK:~$ chmod +x quotes.sh
lenovo@SAISHASHANK:~$ ./quotes.sh
----- SINGLE QUOTES -----
$name
Welcome to Ubuntu
Today is $(date)
Hello          Ubuntu          Linux
----- DOUBLE QUOTES -----
My name is Aruntej
I am studying Computer Science
Current directory: /home/lenovo
Current date: Tue Sep 22 04:38:53 UTC 2026

With double quotes:
<Ubuntu Linux Lab>

Without double quotes:
<Ubuntu>
<Linux>
<Lab>
lenovo@SAISHASHANK:~$
```

```
lenovo@SAISHASHANK:~$ echo "Special characters: $ @ # % & *"
Special characters:  @ # % & *
lenovo@SAISHASHANK:~$ name="Saishashank"
lenovo@SAISHASHANK:~$ echo "$name"
Saishashank
lenovo@SAISHASHANK:~$ printf '<%s>\n' "$name"
<Saishashank>
lenovo@SAISHASHANK:~$ empty=""
lenovo@SAISHASHANK:~$ printf '<%s>\n' "$empty"
<>
lenovo@SAISHASHANK:~$ mkdir "My Ubuntu Project"
lenovo@SAISHASHANK:~$ cd "My Ubuntu Project"
lenovo@SAISHASHANK:~/My Ubuntu Project$ pwd
/home/lenovo/My Ubuntu Project
lenovo@SAISHASHANK:~/My Ubuntu Project$ cd ..
lenovo@SAISHASHANK:~$ rmdir "My Ubuntu Project"
lenovo@SAISHASHANK:~$ nano quotes.sh
lenovo@SAISHASHANK:~$ chmod +x quotes.sh
lenovo@SAISHASHANK:~$ ./quotes.sh
----- SINGLE QUOTES -----
$name
Welcome to Ubuntu
Today is $(date)
Hello          Ubuntu          Linux
----- DOUBLE QUOTES -----
My name is Aruntej
I am studying Computer Science
Current directory: /home/lenovo
Current date: Tue Sep 22 04:38:53 UTC 2026

With double quotes:
<Ubuntu Linux Lab>

Without double quotes:
<Ubuntu>
<Linux>
<Lab>
lenovo@SAISHASHANK:~$
```

```
lenovo@SAISHASHANK: ~$ message -name ts learning ubuntu
lenovo@SAISHASHANK: ~$ echo "$message"
$NAME is learning Ubuntu
lenovo@SAISHASHANK: ~$ set -x
echo '$NAME'
set +x
++ _systemd_osc_context_ps0
++ '[' -n bfa16136-5f5f-42ca-b08f-adf339232281 ']'
+++ _systemd_osc_context_common
+++ '[' -f /etc/machine-id ']'
+++ printf '%machineid=%.36s' 3fecdc9c74f742bda6fc43818cfa767b
+++ printf '%user=%.255s;hostname=%.255s;bootid=%.36s;pid=%.20d' lenovo SAISHASHANK d5d21ee1-1fe4-462d-96df-022bd91e608f 124
0
+++ _systemd_osc_context_escape /home/lenovo
+++ echo /home/lenovo
+++ sed -e 's/\\/\\\\x5c/g' -e 's/;/\\\\x3b/g' -e 's/[[:cntrl:]]/\0/g'
+++ printf '\033]3008;start=.%64s%;type=command;cwd=.%255s\033\ ' bfa16136-5f5f-42ca-b08f-adf339232281 ':machineid=3fecdc9c74f742bda6fc43818cfa767b;user=lenovo;hostname=SAISHASHANK;bootid=d5d21ee1-1fe4-462d-96df-022bd91e608f;pid=00000000000000001240
/home/lenovo
+ echo '$NAME'
$NAME
++ _systemd_osc_context_ps0
++ '[' -n bfa16136-5f5f-42ca-b08f-adf339232281 ']'
+++ _systemd_osc_context_common
+++ '[' -f /etc/machine-id ']'
+++ printf '%machineid=%.36s' 3fecdc9c74f742bda6fc43818cfa767b
+++ printf '%user=%.255s;hostname=%.255s;bootid=%.36s;pid=%.20d' lenovo SAISHASHANK d5d21ee1-1fe4-462d-96df-022bd91e608f 124
0
+++ _systemd_osc_context_escape /home/lenovo
+++ echo /home/lenovo
+++ sed -e 's/\\/\\\\x5c/g' -e 's/;/\\\\x3b/g' -e 's/[[:cntrl:]]/\0/g'
```

```

lenovo@SAISHASHANK: ~$ cd /home/lenovo/Ubuntu Lab
lenovo@SAISHASHANK: /home/lenovo/Ubuntu Lab$ cd ..
lenovo@SAISHASHANK: ~$ text="ubuntu Linux Lab"

printf '%s\n' "$text"
<Ubuntu Linux Lab>
lenovo@SAISHASHANK: ~$ printf '<%s\n' $text
<Ubuntu>
<Linux>
<Lab>
lenovo@SAISHASHANK: ~$ "text"
Ubuntu Linux Lab: command not found
lenovo@SAISHASHANK: ~$ $text
Ubuntu: command not found
lenovo@SAISHASHANK: ~$ set -x
name="Saishashank"
echo "Hello $name"
set +x
++ systemd_osc_context_pso
+++ T [-n 3ed5f991-211d-4330-87b2-949cd751873c '']
+++ T [-f /etc/machine-id '']
+++ printf '%s\n' 'machineId=%x.36s;3fecdc474f742bda6fc43818cfa767b
+++ printf '%user=%x.255s;hostname=%x.255s;bootId=%x.36s;pid=%x.20d' lenovo SAISHASHANK d5d21ee1-1fe4-462d-96df-022bd91e608f 1644
+++ systemd_osc_context_escape /home/lenovo
+++ echo /home/lenovo
+++ sed -e 's/\x3b/g' -e 's/;/\\x3b/g' -e 's/[[:cntrl:]]/\\w/g'
+++ printf '%033j0008;start=%x.645s;type=command;cwd=%x.255s%033j' 3ed5f991-211d-4330-87b2-949cd751873c 'machineId=3fecdc474f742bda6fc43818cfa767b;
df 22bd91e608f;pid=00000000000000001644' /home/lenovo
name="Saishashank"
++ systemd_osc_context_pso
+++ T [-n 3ed5f991-211d-4330-87b2-949cd751873c '']
+++ T [-f /etc/machine-id '']
+++ printf '%s\n' 'machineId=%x.36s;3fecdc474f742bda6fc43818cfa767b
+++ printf '%user=%x.255s;hostname=%x.255s;bootId=%x.36s;pid=%x.20d' lenovo SAISHASHANK d5d21ee1-1fe4-462d-96df-022bd91e608f 1644
+++ systemd_osc_context_escape /home/lenovo
+++ echo /home/lenovo
+++ sed -e 's/\x3b/g' -e 's/;/\\x3b/g' -e 's/[[:cntrl:]]/\\w/g'
+++ printf '%033j0008;start=%x.645s;type=command;cwd=%x.255s%033j' 3ed5f991-211d-4330-87b2-949cd751873c 'machineId=3fecdc474f742bda6fc43818cfa767b;
d5d21ee1-1fe4-462d-96df-022bd91e608f;pid=00000000000000001644' /home/lenovo
++ echo 'Hello Saishashank'
name="Saishashank"
++ systemd_osc_context_pso
+++ T [-n 3ed5f991-211d-4330-87b2-949cd751873c '']
+++ T [-f /etc/machine-id '']
+++ printf '%s\n' 'machineId=%x.36s;3fecdc474f742bda6fc43818cfa767b
+++ printf '%user=%x.255s;hostname=%x.255s;bootId=%x.36s;pid=%x.20d' lenovo SAISHASHANK d5d21ee1-1fe4-462d-96df-022bd91e608f 1644
+++ systemd_osc_context_escape /home/lenovo
+++ echo /home/lenovo

```

```
root@kali:~/Documents# $ set -x
echo Hello Ubuntu
set -e
++ systemd_esc_context_p09
+++ # 7f4bd564-1920-47ee-824f-6bbe3ff96b6d '/'
+++ --systemd_esc_context_common
++++ /etc/machine-id
++++ printf '%m\n%d%k.%3s' 3fecdc474742bda6fc43818cfa767d lenovo SAISWASHANK d5d21ee1-fe4-462d-96df-022bd91eddbf 2211
++++ user=k.255;hostname=k.255;bootid=k.30s;pid=k.20d
+++ --systemd_esc_context_escape /home/lenovo
+++ echo /home/lenovo
+++ sed -e 's/[\\|&$]*' -e 's/[\\|&$]/g' -e 's/[[:cntrl:]]/\[M/g'
+++ [0a33]000;start=k.64sk;type=command;cwd=k.255;(033) 7f4bd564-1920-47ee-824f-6bbe3ff96b6d '[machined:3fecdc474742bda6fc43818cfa767d;user=lenovo;hostname=SAISWASHANK;bootid=d5d21ee1-fe4-462d-96df-022bd91eddbf;pid=889906080606062211]' /home/lenovo
+ echo Hello Ubuntu
--
++ systemd_esc_context_p09
+++ # 7f4bd564-1920-47ee-824f-6bbe3ff96b6d '/'
+++ --systemd_esc_context_common
++++ /etc/machine-id
++++ printf '%m\n%d%k.%3s' 3fecdc474742bda6fc43818cfa767d lenovo SAISWASHANK d5d21ee1-fe4-462d-96df-022bd91eddbf 2211
++++ user=k.255;hostname=k.255;bootid=k.30s;pid=k.20d
+++ --systemd_esc_context_escape /home/lenovo
+++ echo /home/lenovo
+++ sed -e 's/[\\|&$]*' -e 's/[\\|&$]/g' -e 's/[[:cntrl:]]/\[M/g'
+++ [0a33]000;start=k.64sk;type=command;cwd=k.255;(033) 7f4bd564-1920-47ee-824f-6bbe3ff96b6d '[machined:3fecdc474742bda6fc43818cfa767d;user=lenovo;hostname=SAISWASHANK;bootid=d5d21ee1-fe4-462d-96df-022bd91eddbf;pid=889906080606062211]' /home/lenovo
root@kali:~/Documents# $ echo Hello Ubuntu Linux Lab
Hello Ubuntu Linux Lab
set -e
$ echo {HOME} >&/ |& $ echo {HOME} <&'/' |& {test}
$ echo { } {test}
{ } {test}
root@kali:~/Documents# $ echo -e "HelloUbuntu"
Hello
Ubuntu
root@kali:~/Documents# $ echo -e "Name:Utsav Shashank"
Name:Utsav Shashank
echo Hello Ubuntu
ls -ldk
echo /
echo \
echo \\
echo \\Hello\
echo \Ubuntu\
printf "%s\n" Hello Ubuntu
printf "%s\n" Hello Ubuntu
echo Hello
echo \
Name:Utsav Shashank
cat Shashank
echo /
echo \
(Hello)
(Utsav,)
>Hello Ubuntu>
>hello
>Ubuntu>
root@kali:~/Documents# $
```